



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
19-12-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

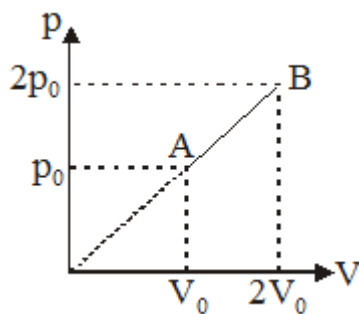
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. An enclosed one mole of a monoatomic gas is taken through a process A to B as shown in figure. The molar heat capacity of the gas for this process is :-



(A) R

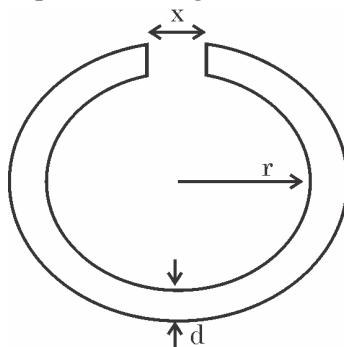
(B) 2R

(C) 3R

(D) 4R

Space for Rough Work

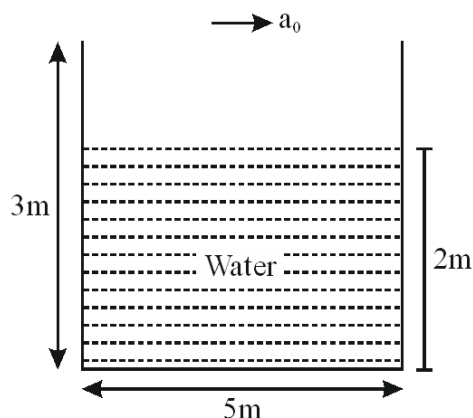
2. A cylindrical metal rod of length L_0 is shaped into a ring with small gap as shown. On heating the system :



- (A) x decreases, r and d increase
 (B) x and r increase, d decrease
 (C) x , r and d all increase
 (D) Data insufficient to arrive at a conclusion.

Space for Rough Work

3. A rectangular container is accelerated horizontally. Water is filled upto a height of 2m in this containers having dimensions $5\text{m} \times 4\text{m} \times 3\text{m}$.



Find maximum acceleration that can be given without spilling the water.

- (A) 4 m/s^2 (B) 3 m/s^2 (C) 5 m/s^2 (D) 7 m/s^2
4. A fixed pulley of radius 20 cm and moment of inertia 0.32 kg.m^2 about its axle has a massless cord wrapped around its rim. A mass M of 2 kg is attached to the end of the cord. The pulley can rotate about its axis without any friction. The acceleration of the mass M is :- (Assume $g = 10 \text{ m/s}^2$)



- (A) 1 m/s^2 (B) 3 m/s^2 (C) 2 m/s^2 (D) 4 m/s^2

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

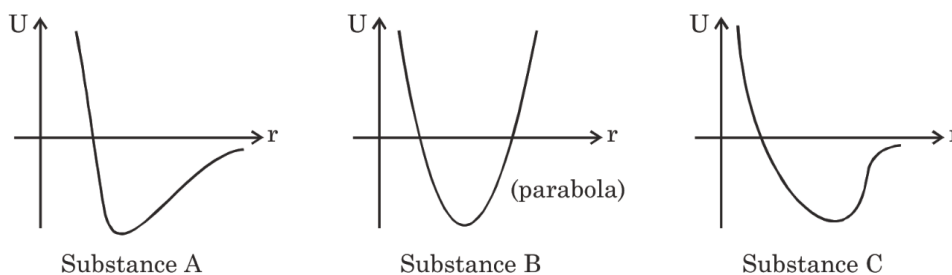
Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
5. A cylindrical empty vessel is kept vertically, with its open end up, on a horizontal ground when it rains uniformly and vertically. The rate of filling in the vessel with rain water will :-
- (A) remain unchanged, if its length is made inclined to the vertical
- (B) decrease, if its length is made inclined with the vertical
- (C) remain unchanged if a slow wind blows horizontally
- (D) decrease, if a slow wind blows horizontally
-

Space for Rough Work

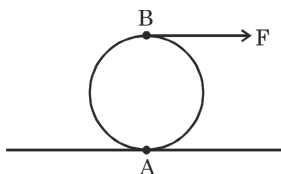
6. Diagram shows the variation of potential energy with separation between the two atoms for three different substances. All the objects are nearly at absolute zero temperature. Choose the correct statement(s) :



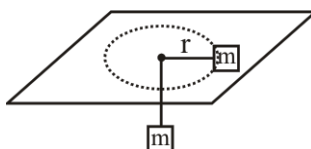
- (A) Substance 'A' expands on heating
- (B) Substance 'B' contracts on heating.
- (C) Substance 'C' contracts on heating.
- (D) Substance 'B' neither expands nor contracts on heating.
7. For an ideal gas :
- (A) the change in internal energy in a constant pressure process from temperature T_1 to T_2 is equal to $nC_V (T_2 - T_1)$, where C_V is the molar specific heat at constant volume and n the number of moles of the gas
- (B) the change in internal energy of the gas and the work done by the gas are equal in magnitude in an adiabatic process
- (C) the internal energy does not change in an isothermal process
- (D) no heat is added or removed in an adiabatic process

Space for Rough Work

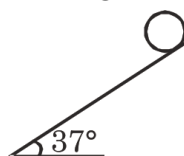
8. A uniform disc is rolling on horizontal surface under the action of force F as shown. Choose the correct statement(s) :-



- (A) Friction force will act towards right.
 (B) Friction force will act towards left.
 (C) Angular momentum of disc is same about point A and B.
 (D) Angular momentum of disc is different about point A and B.
9. In the figure the block on the smooth table is set into motion in a circular orbit of radius r around the centre hole. The hanging mass is identical to the mass on the table and remains in equilibrium. Neglect friction. The string connecting the two blocks is massless and inextensible. Select the correct answer(s).



- (A) The angular speed ω of the block in its circular motion is $\sqrt{\frac{g}{r}}$
 (B) Kinetic energy as function of r is given by $\frac{mgr}{2}$
 (C) Angular momentum about the hole is conserved even if hanging block is pulled down
 (D) The block on table is in equilibrium
10. A solid sphere of mass 1 kg and radius 10 cm is rolling down an inclined plane as shown :-



- (A) It's acceleration is 6 m/s^2
 (B) Its angular acceleration is 300 rad/s^2
 (C) The minimum friction coefficient required for pure rolling is $\frac{3}{14}$.
 (D) Friction is acting downwards.

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

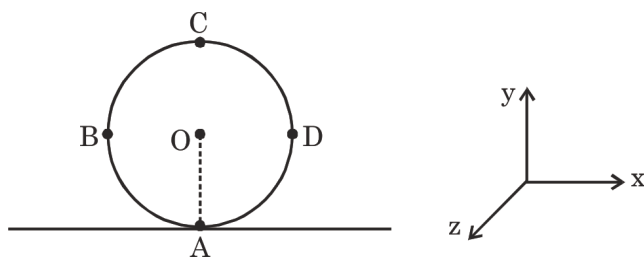
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- There are **TWO (02)** questions corresponding to each question stem.
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- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A uniform ring of radius 1m is performing rolling motion on the horizontal surface as shown in the figure. At $t = 0$ acceleration of COM of ring is $3\hat{i} \text{ m/s}^2$ while angular acceleration & angular velocity are $-3\hat{k} \text{ rad/s}^2$ and $-2\hat{k} \text{ rad/s}$ where x-z plane is horizontal and y is vertical axis. A is in contact with surface and C is at 2m above the surface. B and D are 1m above the surface.

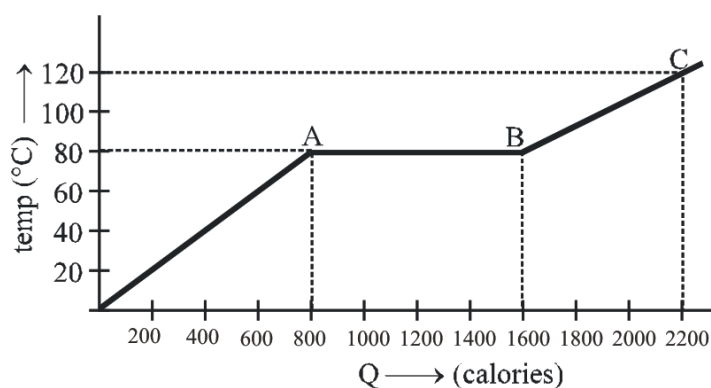


1. Magnitude of acceleration (in m/s^2) of point B with respect to D is :
2. Magnitude of acceleration (in m/s^2) of point A is :

Space for Rough Work

Paragraph for Questions 3 and 4

A substance is in the solid form at 0°C . The amount of heat added to this substance and its temperature are plotted in the following graph. If the relative specific heat capacity of the solid substance is 0.05, find from the graph

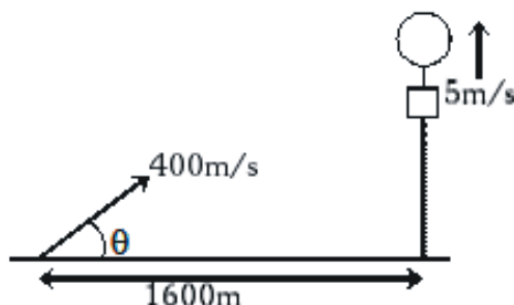


3. The mass of the substance (in kg)
4. The specific latent heat of the melting process (in cal/kg)

Space for Rough Work

Paragraph for Questions 5 and 6

A gun crew observes a remotely controlled balloon launching an instrumented spy package in enemy territory. When first noticed the balloon is at an altitude of 800 m and moving vertically upward at a constant velocity of 5 m/s. It is 1600 m down range. Shells fired from the gun have an initial velocity of 400 m/s at a fixed angle ($\sin\theta = 3/5$ and $\cos\theta = 4/5$). The gun crew waits and fires so as to destroy the balloon. Assume $g = 10 \text{ m/s}^2$. Neglect air resistance.



5. Velocity of shell just before collision in ms^{-1} is (approximately)
6. The altitude of collision in meter is

Space for Rough Work

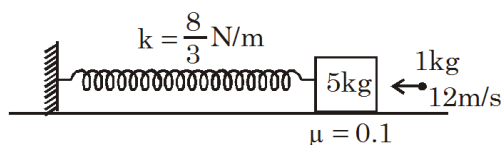
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
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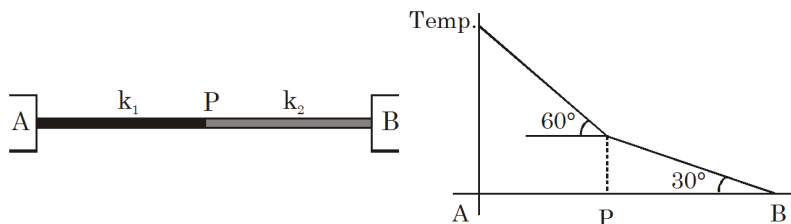
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

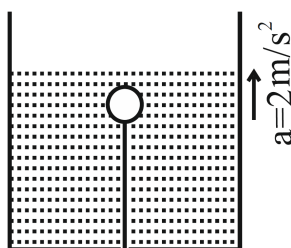
7. A particle of mass 1kg collides and sticks with 5kg block, connected to spring at natural length. If the maximum compression in the spring in subsequent motion is α (in meter). Fill the value of 4α .



8. Two conducting rods are joined at P and end A and end B are kept at fixed temperature. Temperature variation from A to B is given below (k_1 , k_2 conductivity of rods). Find ratio of thermal conductivities $\left(\frac{k_2}{k_1}\right)$ of the rods.



9. A solid sphere of mass $m = 2$ kg and specific gravity $S = 0.5$ is held stationary relative to tank filled with water. The tank is accelerating upward with acceleration 2 m/s^2 . The tension in the string is found to be 8 n . Find the value of n . Given density of water $= 1000 \text{ kg m}^{-3}$ and $g = 10 \text{ m/s}^2$.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

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1. The hemoglobin of most mammals contains approximately 0.33% of iron by mass. The molecular mass of hemoglobin is 67200. The number of iron atoms in each molecule of hemoglobin is (Fe = 56)
(A) 3 (B) 4 (C) 2 (D) 6
2. In the electronic configuration of Zn^{+2} , find total electrons present in those orbitals which have no nodal plane.
(A) 8 (B) 10 (C) 6 (D) 4
3. In the following reaction :

$$\text{NO}_2^+ + 2\text{e}^- \rightarrow \text{NO}_2^-$$
 Select the incorrect statement :-
 (A) Hybridisation of central atom gets changed
 (B) Formal charge on central atom gets changed
 (C) Oxidation state of central atom gets changed
 (D) Maximum number of atoms lying in one plane gets changed
4. Compound of molecular formula $\text{C}_4\text{H}_8\text{O}$ may have functional group -
 (a) Alcohol (b) Ether (c) Carboxylic acid (d) Aldehyde
 (A) a,b,c (B) b,c & d (C) a,b & d (D) a,b,c & d

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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5. Which of the following statements is(are) true ?

(A) $\frac{(U_{rms})_{O_2,TK}}{(U_{av})_{O_2,TK}}$, is constant, independent of T

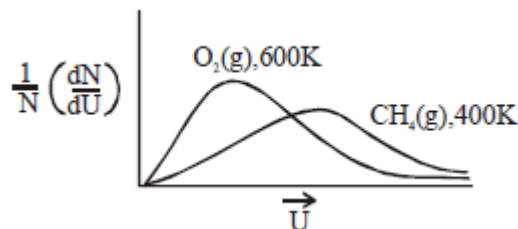
(B) Wavelength of ejected photoelectron can be calculated by $\lambda = \frac{hc}{E}$, where E is kinetic energy of ejected photo electron.

(C) Ozone gas can be absorbed by alkaline pyrogallol solution

(D) Oxygen gas can be absorbed by alkaline pyrogallol solution.

Space for Rough Work

6. Correct statement(s) for the ideal gas from the given graph is/are-

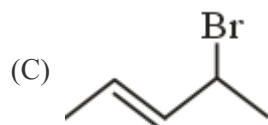
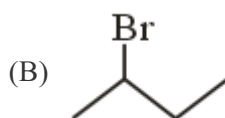
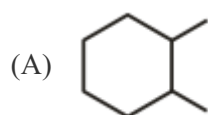


- (A) Average speed of $\text{CH}_4(\text{g})$ is higher than $\text{O}_2(\text{g})$.
 (B) Average kinetic energy per mole of $\text{CH}_4(\text{g})$ is higher than that of $\text{O}_2(\text{g})$.
 (C) Fraction of molecules showing most probable speed is higher for $\text{O}_2(\text{g})$
 (D) Most probable speed of $\text{O}_2(\text{g})$ is higher than $\text{CH}_4(\text{g})$
7. Choose the **INCORRECT** match in the following indicated properties.
- (A) N_2O_5 : Most acidic oxide among the P_4O_{10} , N_2O_5 , BeO , SO_3 , SiO_2 , Cl_2O_7
 (B) Cl : Most electron affinity among the F , Na^+ , F^+ , Cl
 (C) Al^{3+} : Smallest cation among the Al^{3+} , Li^+ & Mg^{2+}
 (D) F^- : Highest ionic mobility in aqueous solution among all halide ions
8. Which of the following is/are species contains at least 2π -bonds?

- (A) IO_2F_2^- (B) $\text{H}_2\text{S}_2\text{O}_3$ (C) HN_3 (D) P_4O_6

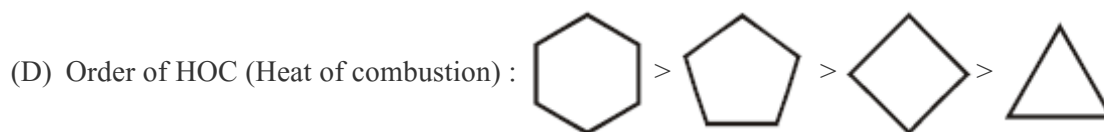
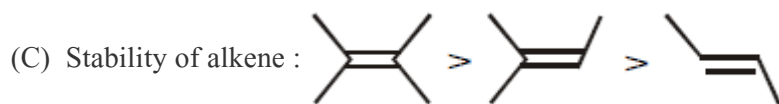
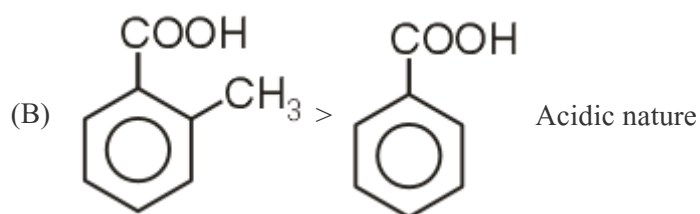
Space for Rough Work

9. Which of the following will show optical as well as Geometrical isomerism



10. Correct options amongst the following is/are :

(A) $\text{Et-NH}_2 > \text{Et-OH}$: Base nature



Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

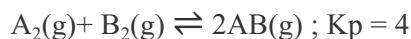
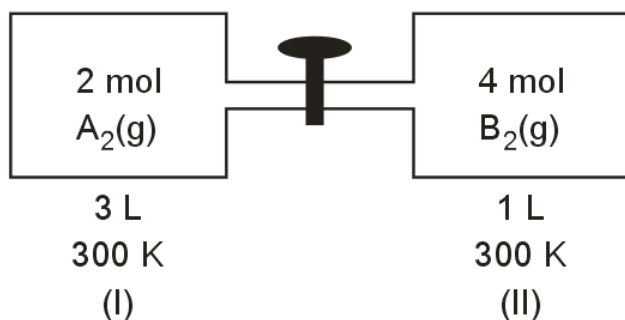
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- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Gases A_2 & B_2 are present in two containers [I & II] respectively at 300K separated by a narrow tube of negligible volume closed by a valve. On opening the valve reaction given below attains equilibrium :

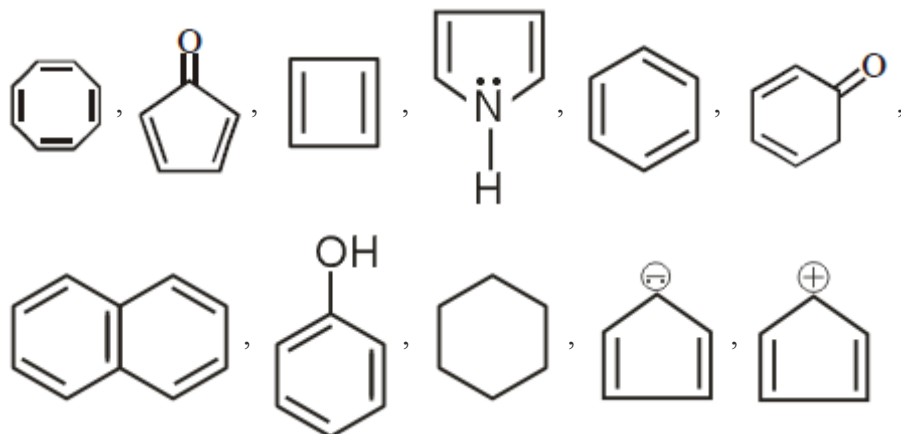


1. Equilibrium molar concentration of $AB(g)$ is -
2. Final moles of $B_2(g)$ in container (I) would be -

Space for Rough Work

Paragraph for Questions 3 and 4

Consider the following structures and answer the given questions



3. Total number of anti aromatic compound(s) -
4. Total number of aromatic compound(s)-

Space for Rough Work

Paragraph for Questions 5 and 6

Information-1: Principal Quantum No. 'n' is defined as 1, 2, 3.....

Information-2: Azimuthal Quantum No. 'ℓ' is defined as 0 to (n + 2) in integral step

Information-3: Magnetic Quantum No. 'm' is defined as $-(\ell+1)$ to $+(\ell+1)$

Information-4: Spin Quantum No. 's' has six possible values $\left(-2, -1, -\frac{1}{2}, +\frac{1}{2}, +1, +2\right)$

Information-5: The sub-shell corresponding to $\ell = 0, 1, 2, 3, 4, 5, \dots$ are designated as F, G, H, I, J, K, respectively.

Information-6: The values of m for a given value of ℓ give the number of orbitals in a sub-shell.

Information-7: The principles for filling electrons in the shells remain unchanged.

On the basis of above information, answer the following questions.

5. Atomic number of last element of first period would be:
6. The maximum number of electrons that can be filled in a I-sub-shell are respectively -

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
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- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. For a certain orbital

$$\Psi(r) = \frac{1}{16\sqrt{4}a_0^{3/2}} (1 - \sigma)(\sigma^2 - 8\sigma + 12) e^{-\sigma/2}, \sigma = \frac{2Zr}{na_0} \{a_0 = \text{constant}\}$$

The distance of nearest radial node from nucleus is xa_0 while farthest radial node from nucleus is ya_0 .
Find $\left(\frac{y}{x}\right)$.

8. Find the sum of d-orbitals which are used in bonding in the given species?
 SO_2 , SO_3 , XeF_2 , H_3PO_3 , SO_3^{2-} , IO_6^{5-}
9. Identify total number of structure of **ether** possible for $\text{C}_4\text{H}_{10}\text{O}$:

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

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- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Words are formed using all the letters of the word MRIDUL. These words are arranged as in a dictionary. Find the rank of word MRIDUL in this dictionary.
(A) 440 (B) 415 (C) 417 (D) 409
2. If $a^2 + b^2 + c^2 + ab + bc + ca \leq 0 \forall a, b, c \in \mathbb{R}$, then the value of determinant $\begin{vmatrix} 1 & a & b \\ c & 1 & a \\ b & c & 1 \end{vmatrix}$ is
(A) abc (B) $1 + a + b + c$ (C) $a + b + c$ (D) $a + b + c + abc$
3. For real numbers a, b, c and d if $a^2 + b^2 = 4$ and $c^2 + d^2 = 1$, then the greatest value of $(ac + bd)$ is
(A) 2 (B) 1 (C) $\frac{1}{4}$ (D) 0
4. Number of ways in which 9 persons can be seated in 3 different taxis each having 3 seats for passengers if internal arrangement of persons inside the taxi is immaterial, is
(A) $\frac{9!}{3! 3!}$ (B) $\frac{9!}{3! 3! 3!}$ (C) $\frac{9!}{3!}$ (D) $9!$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. If the coefficient of x^m , $m \in \mathbb{N}$ in the expansion of $\left(\sqrt{2}x + \sqrt[3]{\frac{3}{x^2}}\right)^9$ is equal to k ($k \neq 0$), then k is equal to -
- (A) 1008 (B) 2016 (C) 3024 (D) 1016
6. If the normal at point P to the rectangular hyperbola $x^2 - y^2 = 4$ meets the transversal and conjugate axes at A and B respectively and C is the centre of the hyperbola, then -
- (A) $PA = PC$ (B) $PA = PB$ (C) $PB = PC$ (D) $AB = 2PC$

Space for Rough Work

7. If one of the foci of an ellipse is $(2, 3)$ whose eccentricity is $\frac{1}{3}$ and point of intersection of its major axis with the corresponding directrix is $(10, 11)$, then identify the correct statement -
- (A) vertices of the ellipse are $(-2, -1)$ and $(4, 5)$
- (B) major axis is $6\sqrt{2}$
- (C) minor axis is 8
- (D) latus rectum is $\frac{32\sqrt{2}}{3}$
8. A normal to parabola $y^2 = 8x$ at P meets the curves again at Q such that the angle subtended by PQ at the vertex is $\frac{\pi}{2}$, then P can be -
- (A) $(16, 8\sqrt{2})$ (B) $(16, 8)$ (C) $(4, -4\sqrt{2})$ (D) $(4, 4\sqrt{2})$
9. For the number $N = 16200$, identify correct statement(s)
- (A) Number of divisors is 60
- (B) Number of divisors which end with 0 is 30
- (C) Number of divisors NOT divisible by 15, is 28
- (D) Number of even divisors is 30
10. Each side of a square is of length 4 unit. The center of the square is $(3, 7)$ and one of its diagonal is parallel to $y = x$. Then identify the vertices of square -
- (A) $(1, 5)$ (B) $(1, 9)$ (C) $(5, 9)$ (D) $(5, 5)$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

If the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be such that one of its focus lie on the line $2x + y = 12$ and there exist only one tangent at a distance of 1 unit from this focus.

1. The length of transverse axis of the hyperbola is -
2. The eccentricity of the hyperbola is -

Space for Rough Work

Paragraph for Questions 3 and 4

Consider the conic $\sqrt{(x+4)^2 + (y-1)^2} + \sqrt{(x-4)^2 + (y-1)^2} = 16$.

3. Perpendicular distance between directrices will be -
4. Distance between any focus and any end of minor axis will be -

Paragraph for Questions 5 and 6

If $f(x) = |x-3| + 2|x-1|$ and $g(x) = -2x^2 + 3x - 1$.

5. Minimum value of $f(x)$ is -
6. Number of solution(s) of the equation $f(x) = g(x)$ is -

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. If the radical axis of the circles $x^2 + y^2 + 2x + 2ky + 5 = 0$ and $x^2 + y^2 + 2kx + k = 0$ is orthogonal to the circle $x^2 + y^2 - 2x - 2y = 0$, then the value of k is
 8. Sum of the series $\frac{3}{1^2 \cdot 2^2} + \frac{5}{2^2 \cdot 3^2} + \frac{7}{3^2 \cdot 4^2} + \dots$ up to infinite terms is equal to
 9. Number of solution of the equation $\cos^4 x - \sin^6 x = 1$ in $[0, 2\pi]$
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work